

Top view drawing of a rectangular plate. The overall dimensions are 52.70 (width) and 90.80 (height). The inner dimensions are 44.80 ± 0.05 (width) and 82.90 ± 0.05 (height). The plate features four corner mounting holes (S 2.0, N 2.0, E 2.0, W 2.5) and two central locator pillars (2x LOCATOR PILLAR Ø3.0 x 0.4 TALL (NOTE 4)). A U2 RELIEF POCKET (NOTE 7) is also indicated. The drawing includes dimension lines and callouts for various features.

SECTION A-A → edge profile DETAIL B → corner boss

SECTION A-A (edge) SCALE 13:1

Technical drawing of a mechanical part showing a cross-section. The part has a central vertical section and two side sections. The central section has a diameter of $\varnothing 5.20$ and a length of $\varnothing 3.0$ for the spotface (back). The side sections have a diameter of $\varnothing 1.6$ and a length of $\varnothing 1.6$. The part is made of a material with a hatched pattern. The drawing includes the following annotations:

- M2 THREAD $\varnothing 1.6$ TAP DRILL, THRU TAP FROM BACK
- spotface bottom = M2x3 screw-tip plane (flush)
- $\varnothing 3.0$ SPOTFACE (BACK)
- $\varnothing 5.20$

1. MATERIAL: TITANIUM Gr5 (TC4) = Ti-6Al-4V GRADE 5.
2. FINISH: BEAD-BLAST MATTE (UNIFORM ON THE STEPPED BACK FACE). REAR ART LASER-MARKED IN THE RECESSED BACK FIELD.
3. GENERAL TOLERANCE PER ISO 2768-1 (MEDIUM). C1-C4 TOLERANCED AS LISTED; C1 & C3 = ± 0.05 . DATUMS: A = LEFT EDGE, B = BOTTOM EDGE, C = PCB-REST PLANE.
4. 2x LOCATOR PILLAR: $\varnothing 3.0 \times 0.4$ TALL, STANDING ON THE CAVITY FLOOR AT (13, 35) AND (33, 55) - BOTH WEST OF THE NFC COIL. LEFT AS ISLANDS IN THE SAME CAVITY PASS AS THE BOSSES.
LOCATE THE RESIN BRACE (SEPARATE PART) VIA MATCHING $\varnothing 3.2$ RECESSES. FLOOR STAYS A FULL 1.00 (THE (33,55) RECESS IS SLOTTED). MODELED IN THE STEP.
5. BREAK ALL SHARP EDGES ~ 0.1 (Ti). OUTER TOP/BOTTOM RIM EASED $0.20 \times 45^\circ$ (MODELED).
6. FLOOR IS NOW A TRUE 1.00 (0.95 LOCAL UNDER THE U2 POCKET), AT THE ~ 1.0 Ti MIN-WALL GUIDANCE SO IT ALSO CLEARS ALUMINIUM / COPPER / STAINLESS.
CAVITY $1.80 + 0.05 \rightarrow 1.75$ WORST-CASE, MINUS WS17 1.70 MAX (DATASHEET CASE WS17 HEIGHT) = 0.05 NON-CONTACT; THE BRACE + SOLAR-CELL SANDWICHES CARRY THE BOARD.
7. U2 RELIEF POCKET: 7.8×5.4 CENTERED (28.5, 37) ON THE CAVITY FLOOR, 0.05 DEEP (LOCAL FLOOR 0.95) - U2 (1.75) KEEPS 0.10 AIR. GENERAL FLOOR 1.00. MODELED IN THE STEP.
8. ASYMMETRIC SUPPORT LIP (SEE PLAN): WEST 2.5 / NORTH 2.0 / SOUTH 2.0 FOR PCB RIGIDITY; EAST 1.0 THROUGH THE JP1/TP1 PADS + OVER THE NFC COIL (A GROUNDED LIP WOULD DETUNE IT), WIDENING TO 2.5 AT THE N/S ENDS. BACK-FRAME MIRRORS IT.
9. NO INTERNAL RIBS OR SUPPORT POSTS: THE RESIN BRACE (SEPARATE PART) CARRIES CENTER SUPPORT. A PCB LAYOUT CHANGE = A BRACE REPRINT, NOT A SHELL RE-MACHINE.
10. PCB RECESS = 0.60 DEEP (RECEIVES THE 0.60 mm BOARD; SLIP FIT, NOT A PRESS FIT). 3D STEP GOVERNS ALL GEOMETRY; STL NOT FOR CNC.
11. PART = BACK-SHELL ONLY PCB, RESIN BRACE, AND 4x M2x3 BRASS SLOTTED-CHEESE SCREWS (HEAD $\varnothing 3.8$) SUPPLIED SEPARATELY.

SOLAR-GLOW DRH v3.0 — Ti BACK-SHELL (0.6mm-BOARD DUMB BOX)	
DWG solar-glow-drh-v3_0-backshell-0p6b-brace	
REV v3.0	
MATERIAL Ti Gr5 (TC4)	
FINISH BEAD-BLAST	
UNITS mm	SCALE AS NOTED
BBOX 52.70 x 90.80 x 3.55 3rd-angle	SHEET 1/1